UNIT 3 PRACTICE ASSESSMENT: INVESTIGATING DATA & SCATTER PLOTS

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Name: _____ **Date:** _____

Part 1: Multiple Choice & Foundational Concepts

1. Scatter plots are graphs that use points to display numerical data with two variables, commonly known as:

A. Categorical data
B. Bivariate data
C. Univariate data
D. Qualitative data

2. When one variable increases while the other variable also consistently increases, what type of association exists?

A. Positive association
B. Negative association
C. No association
D. Constant association

3. When one variable increases while the second variable decreases, what type of association is shown?

A. Positive linear association
B. Negative association
C. Non-linear constant
D. No association

4. A single data point that lies distinctly away from the overall pattern or trend of a scatter plot is termed:

A. A cluster
B. The origin
C. An outlier
D. An intercept

5. When data points are spread completely across the grid with no discernible upward or downward direction:

A. Strong positive association
B. Strong negative association
C. Weak nonlinear association
D. No association

6. If the points in a scatter plot lie very tightly along an approximate straight line path, the association is:

A. Weak and non-linear
B. Strong and linear
C. Weak and linear
D. Strong and non-linear

7. When data points form distinct, isolated groups on a scatter plot separated by noticeable gaps, this displays:

A. Outliers
B. Curvature
C. Clustering
D. Negative slope

Part 2: Real-World Associations

State whether each situation would show a **positive**, **negative**, or **no association**.

8. The number of hours a basketball player practices shooting and their field goal percentage.

Association: _____

9. A person's adult shoe size and their score on a high school mathematics examination.

Association: _____

10. The number of highway miles driven in an automobile and the gallons of gasoline remaining in the fuel tank.

Association: _____

11. The daily high outdoor temperature and the total quantity of hot cocoa sold at a mountain lodge.

Association: _____

12. The square footage of an apartment and the monthly rent charged for that apartment.

Association: _____

Part 3: Data Analysis & Interpretation

Data Set 1: Distance vs. Baskets Made (Students shot for 30 seconds from designated distances):

Student	Distance from Hoop (ft)	Baskets Made	Student	Distance from Hoop (ft)	Baskets Made
Leo	10	9	Dave	30	4
Sam	10	8	Eric	35	3
Maya	15	2	Frank	40	1
Ben	15	8	Grace	45	2
Chloe	30	5	(End of recording)		

13. Describe the type of association (positive/negative/none) between shooting distance and baskets made. Explain what this means in words.

14. Identify the coordinate point (x, y) that represents an outlier in this data set. Which student is this, and why is it an outlier?

15. Does the data demonstrate clustering? Explain by referencing the specific distance groupings observed on the court.

Data Set 2: City Population & Parks (Towns surveyed across a county):

Town	Population (thousands)	Parks	Town	Population (thousands)	Parks
Town A	15	4	Town E	65	15
Town B	20	5	Town F	75	18
Town C	35	8	Town G	25	17
Town D	50	12	Town H	85	21

16. Describe the association between population size and the number of public parks (direction and linear form).

17. Which town's data point represents an outlier? Explain using the expected trend for that population.

Data Set 3: Lunch Account Balances (Student spending recorded over time):

Days Since Deposit	2	4	7	10	14	16	18
Balance Remaining ($)	19.00	16.50	13.00	9.50	5.25	3.50	1.75

18. State whether this relationship is positive or negative, and explain what happens to the dollar balance as days increase.

19. Does this data set contain any obvious outliers? Explain your answer.

Part 4: Extended Reasoning

20. Marcus surveys classmates on weekly chores and weekly allowance. He observes that students who complete fewer chores receive less allowance. Marcus states: *"Because both variables are decreasing, the association is negative."*
Is Marcus correct? Explain why or why not using the mathematical definitions of positive and negative associations.

Teacher Scoring Guide & Answer Key

1. B (Bivariate data)

2. A (Positive association)

3. B (Negative association)

4. C (An outlier)

5. D (No association)

6. B (Strong and linear)

7. C (Clustering)

8. Positive association (practice improves performance)

9. No association (shoe size does not affect math skill)

10. Negative association (more miles driven = less gas)

11. Negative association (hotter temperature = less hot cocoa)

12. Positive association (larger space = higher rent)

13. Negative linear; as shooting distance increases, baskets made decreases.

14. Maya (15, 2); at 15 ft, other students scored around 8, making 2 an anomaly.

15. Yes; two distinct groups: short distance (10–15 ft) and long distance (30–45 ft) with a gap between 15–30 ft.

16. Positive and linear association (larger population relates to more parks).

17. Town G (25, 17); for 25k population, ~6 parks would be expected, but it has 17.

18. Negative; as the number of elapsed days increases, the money remaining in the account decreases.

19. No outliers; every data point closely follows the steady downward rate of spending.

20. Incorrect. A positive association means variables move in the same direction (both increase or both decrease). Negative means opposite directions.